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METHOD FOR PRODUCING TUBE BODY

Abstract

A tube body production method includes repeating a fiber layer forming step to form fiber layers so that the fiber layers are disposed in a manner of being stacked in a radial direction of a cylindrical tube member. Each fiber layer forming step includes, to form a current fiber layer: disposing a respective fiber body with respect to an outer circumferential surface of the cylindrical tube member so that the respective fiber body extends in an axial direction of the cylindrical tube member; and winding a respective first fixing member with respect to the outer circumferential surface of the cylindrical tube member such that the respective first fixing member is wound over the respective fiber body by one or more turns in a circumferential direction, along the axial direction of the cylindrical tube member, at a different phase with respect to the first fixing members disposed for other fiber layers.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a divisional application of the U.S. patent application Ser. No. 17/884,215, filed Aug. 9, 2022, which claims priority to PCT/JP2020/012536 filed on Mar. 19, 2020, the disclosures of which are hereby incorporated by reference in their entireties.

TECHNICAL FIELD

[0002] The present invention relates to an intermediate of a tube body which is, for example, used as a power transmission shaft of a vehicle and to a method of producing a tube body.

BACKGROUND ART

[0003] Power transmission shafts (propeller shafts) mounted on vehicles each include a tube body extending in the front-rear direction of the vehicle. The tube body transmits power generated by a power plant and decelerated by a transmission to a final reduction gear unit. There is known a tube body made of a fiber reinforced plastic and produced using a mandrel, as a tube body used for such a power transmission shaft (see Japanese Patent Publication No. H03-265738 A).

SUMMARY OF THE INVENTION

Problems to be Solved by the Invention

[0004] Known techniques for winding a material over a mandrel includes the filament winding technique, which involves winding a continuous fiber impregnated with a resin, and the sheet winding technique, which involves winding a prepreg (a sheet made by impregnating a fiber material with a resin). The filament winding technique is advantageous for low-cost production but has difficulty in aligning a fiber material along the axial direction of a mandrel, in other words, aligning a fiber material along the axial direction of the tube body. On the other hand, the sheet winding technique allows aligning a fiber material along the axial direction of a mandrel but is disadvantageous in terms of production cost.

[0005] Here, assume that L represents the axial direction length over which fibers are disposed over a mandrel, r represents the radius of the outer circumferential surface of the mandrel, and θ represents the orientation angle of the fibers with respect to the mandrel (see FIG. 1). When the orientation angle θ is small such that $\tan \theta < |2\pi r/L|$, the fiber fails to be wound around the mandrel by one or more turns, in which case gravity can cause the fibers to separate from the mandrel.

[0006] The present invention is made to resolve such a problem, and it is an object of the present invention to provide a tube body intermediate and a tube body production method with which it is possible to, while reducing the production cost, reduce displacement of a fiber body even when the orientation angle of the fiber body is small

Means for Solving the Problems

[0007] To resolve the above-described problems, an aspect of the present invention is a tube body production method including: [0008] a step of repeating a fiber layer forming step for a plurality of times to form a plurality of fiber layers so that the plurality of fiber layers are disposed in a manner of being stacked in a radial direction of a circular cylindrical tube member, each fiber layer forming step comprising, to form a respective fiber layer comprised of a respective fiber body and a

respective first fixing member, [0009] a disposing step of disposing the respective fiber body with respect to an outer circumferential surface of the circular cylindrical tube member so that the respective fiber body extends in an axial direction of the circular cylindrical tube member, and [0010] a fixation step of winding the respective first fixing member, which is configured to fix the respective fiber body with respect to the outer circumferential surface of the circular cylindrical tube member, with respect to the outer circumferential surface of the circular cylindrical tube member such that the respective first fixing member is wound over the respective fiber body by one or more turns in a circumferential direction, along the axial direction of the circular cylindrical tube member; and [0011] a molding step of, after the step of repeating the fiber layer forming step, impregnating the fiber bodies of the plurality of fiber layers with a resin on the outer circumferential surface of the circular cylindrical tube member and then heating the resin to mold the resin, [0012] wherein the first fixing member of a lower one of the plurality of fiber layers and the fiber body of an upper one of the plurality of fiber layers, located on top of the lower one, are at least stacked in the radial direction, and [0013] wherein each of the repeated fixation steps comprises disposing the first fixing member of the corresponding fiber layer at a different phase with respect to the first fixing members disposed in the other fixation steps in the axial direction of the circular cylindrical tube member.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0014] FIG. 1 is a diagram schematically illustrating a tube body intermediate according to a first embodiment of the present invention.

[0015] FIG. 2 is a diagram schematically illustrating a mandrel according to the first embodiment of the present invention and a first carbon fiber layer of the tube body intermediate according to the first embodiment of the present invention.

[0016] FIG. 3 is a diagram schematically illustrating a second carbon fiber layer of the tube body intermediate according to the first embodiment of the present invention.

[0017] FIG. 4 is a diagram schematically illustrating a third carbon fiber layer of the tube body intermediate according to the first embodiment of the present invention.

[0018] FIG. 5 is a diagram schematically illustrating a molding device for producing the tube body according to the first embodiment of the present invention.

[0019] FIG. 6 is a diagram schematically illustrating a tube body produced using the tube body intermediate according to the first embodiment of the present invention.

[0020] FIG. 7 is a flowchart illustrating a tube body production method according to the first embodiment of the present invention.

[0021] FIG. 8 is a diagram schematically illustrating an example of the tube body according to the first embodiment of the present invention.

[0022] FIG. 9 is a cross-sectional view schematically illustrating a tube body intermediate according to a second embodiment of the present invention.

[0023] FIG. 10 is a flowchart illustrating a tube body production method according to the second embodiment of the present invention.

[0024] FIGS. 11A to 11D is explanatory diagrams schematically illustrating the tube body intermediate production method according to the second embodiment of the present invention.

[0025] FIG. 12 is a diagram schematically illustrating a tube body produced using the tube body intermediate according to the second embodiment of the present invention.

[0026] FIG. 13 is a diagram schematically illustrating a tube body intermediate according to a third embodiment of the present invention.

[0027] FIG. 14 is a flowchart illustrating a tube body production method according to the third

embodiment of the present invention.

MODES FOR CARRYING OUT THE INVENTION

[0028] An embodiment of the present invention will be described in detail with reference to the drawings, taking an exemplary case of producing a power transmission shaft (propeller shaft) of a vehicle as a tube body using a carbon fiber reinforced plastic. In the descriptions given below, the same elements are denoted by the same reference signs, and overlapping descriptions will be omitted. The drawings referenced in the description are depicted in a deformed manner for ease of understanding such that the dimensions of the members (e.g., shape or the like of the joints **22** and **23** illustrated in FIGS. 5 and 7) are not represented accurately.

First Embodiment

[0029] The tube body intermediate **10A** illustrated in FIG. 1 is formed by disposing carbon fiber layers on the outer circumferential surface of a mandrel **1** (see FIG. 2). As illustrated in FIG. 2, the mandrel **1** is a metal member having a circular cylindrical tube shape.

Tube Body Intermediate

[0030] As illustrated in FIG. 1, the tube body intermediate **10A** according to the first embodiment of the present invention is a circular cylindrical tube member which has a plurality of carbon fiber layers stacked one above the other and which is formed in the middle of producing a tube body **20A** (see FIG. 6) described later. The tube body intermediate **10A** includes, in order from the radially inner side (from the mandrel **1** side), a first carbon fiber layer **11** (see FIG. 2), a second carbon fiber layer **12** (see FIG. 3), and a third carbon fiber layer **13** (see FIG. 4). The first carbon fiber layer **11**, the second carbon fiber layer **12**, and the third carbon fiber layer **13** constitute a reinforced carbon layer to be included in the resin **21** of the later-described tube body **20** to reinforce the resin **21**. The tube body intermediate **10A** further includes a first fixing member **13b** as a member for fixing carbon fibers **13a** in the third carbon fiber layer **13**. Note that, FIGS. 2 to 4 only partially illustrate the carbon fiber layers **11**, **12**, and **13**.

<<First Carbon Fiber Layer>>

[0031] As illustrated in FIG. 2, the first carbon fiber layer **11** is constituted by a plurality of carbon fibers **11a** disposed with respect to the outer circumferential surface of the mandrel **1** so as to cover the mandrel **1**. The carbon fibers **11a** of the first carbon fiber layer **11** are disposed such that the carbon fibers **11a** are wound by one or more turns so as to be inclined by 45 degrees with respect to the axial direction of the mandrel **1** and that the carbon fibers **11a** extend helically with respect to the axial direction of the mandrel **1**. In other words, the orientation angle θ of the carbon fibers **11a** is 45 degrees with respect to the axis X of the mandrel **1**.

<<Second Carbon Fiber Layer>>

[0032] As illustrated in FIG. 3, the second carbon fiber layer **12** is disposed on the radially outer side of the first carbon fiber layer **11** and is constituted by a plurality of carbon fibers **12a** disposed so as to cover the first carbon fiber layer **11**. The carbon fibers **12a** of the second carbon fiber layer **12** are disposed such that the carbon fibers **12a** are wound by one or more turns so as to be inclined by -45 degrees with respect to the axial direction of the mandrel **1** and that the carbon fibers **12a** extend helically with respect to the axial direction of the mandrel **1**. In other words, the orientation angle θ of the carbon fibers **12a** is -45 degrees with respect to the axis X of the mandrel **1**.

<<Third Carbon Fiber Layer>>

[0033] As illustrated in FIG. 4, the third carbon fiber layer **13** is disposed on the radially outer side of the second carbon fiber layer **12** and is constituted by: a plurality of carbon fibers **13a** disposed so as to cover the second carbon fiber layer **12**; and the first fixing member **13b**. The carbon fibers **13a** of the third carbon fiber layer **13** are disposed to extend in parallel with the axial direction of the mandrel **1**. In other words, the orientation angle θ of the carbon fibers **13a** with respect to the axis X of the mandrel **1** is 0 degrees. The carbon fibers **13a** has a length equal to the axial direction length L of a portion of the mandrel **1** on which portion the carbon fiber layers **11** to **13** are disposed. Note that the mandrel **1** has opposite end portions which are to be held by devices and

thus on which the carbon fiber layers **11** to **14** are not disposed.

<<First Fixing Member>>

[0034] As illustrated in FIG. **1**, the first fixing member **13b** is a member configured to fix the carbon fibers **13a** on the outer circumferential surface of the mandrel **1**. The first fixing member **13b** is a flexible resin member having, for example, a thread-like or belt-like shape. The first fixing member **13b** may be formed of the same material as the later-described resin **21**. The first fixing member **13b** may be formed of a material that melts due to the heat(s) of a molding device (mold) **2** and/or of the resin **21** to mix into the resin **21**. The first fixing member **13b** is disposed such that the first fixing member **13b** is wound by one or more turns so as to be inclined with respect to the axial direction of the mandrel **1** and that the first fixing member **13b** extends helically with respect to the axial direction of the mandrel **1**.

[0035] The first fixing member **13b** as described above prevents the carbon fibers **13a**, which are disposed on the outer circumferential surface of the mandrel **1** placed such that the axial direction thereof extends in the horizontal direction (i.e., which are disposed on the outer circumferential surface of the horizontally placed mandrel **1**), from hanging down due to gravity. Specifically, the first fixing member **13b** suitably prevents, of a portion of carbon fibers **13a** located on a lower portion of the outer circumferential surface of the mandrel **1**, an axial direction intermediate portion from hanging down due to gravity.

Tube Body Production Method

[0036] Subsequently, a description will be given of a method of producing a tube body **20A** (see FIG. **6**) via the tube body intermediate **10A** according to the first embodiment, using the flowchart illustrated in FIG. **7**.

[0037] Firstly, a joint (stub yoke or stub shaft) **22** (see FIG. **5**) is disposed (step **S1**: Joint installation step) to one axial direction end portion of the mandrel **1**. Subsequently, as illustrated in FIG. **2**, the first carbon fiber layer **11** is formed (step **S2**: First carbon fiber layer forming step) on the outer circumferential surface of the mandrel **1** by a not-illustrated device. Subsequently, as illustrated in FIG. **3**, the second carbon fiber layer **12** is formed (step **S3**: Second carbon fiber layer forming step) on the outer circumferential surface of the first carbon fiber layer **11** by a not-illustrated device. Subsequently, as illustrated in FIG. **4**, carbon fibers **13a** of the third carbon fiber layer **13** are formed (step **S4**: Third carbon fiber layer forming step/Disposing step) on the outer circumferential surface of the second carbon fiber layer **12** by a not-illustrated device.

[0038] Subsequently, as illustrated in FIG. **1**, by a not-illustrated device, the first fixing member **13b** is disposed on the outer circumferential surface of the carbon fibers **13a** of the third carbon fiber layer **13**, thereby the carbon fibers **13a** are fixed (step **S5**: Third carbon fiber layer forming step, Fixation step) with respect to the mandrel **1**.

[0039] The above-described steps from the first carbon fiber forming step to the fixation step can be said as a tube body intermediate production method configured to produce the tube body intermediate **10A**.

[0040] Subsequently, as illustrated in FIG. **5**, the first carbon fiber layer **11**, the second carbon fiber layer **12**, and the third carbon fiber layer **13** are impregnated with a resin **21** by a molding device (mold) **2**. Then, heat is applied to the molding device **2** to form (step **S6**: Molding step) the tube body **20**. The resin **21** is, for example, a thermosetting resin. In the case of the present embodiment, the mold of the molding device **2** is divided into plural parts. In the molding step, while applying heat to the tube body intermediate **10A**, after a mold closing operation that closes the mold of the molding device **2**, a mold clamping operation that applies pressure to the closed mold is performed to increase the pressure in the mold, thereby to facilitate curing of the resin **21**. Note that, as the present embodiment is described as using a mold having a plurality of parts, the mold closing operation and the mold clamping operation are described as being performed. However, the mold clamping operation is not necessarily mandatory. In addition, when the mold is not divided into a plurality of parts, such mold closing operation and mold clamping operation are not necessarily

mandatory. In the example illustrated in FIG. 5, the joint (stub yoke or stub shaft) **22** is disposed to the one axial direction end portion of the mandrel **1**, and the tube body intermediate **10A** extends to an outer circumferential surface of the joint **22**. Moreover, in the molding device **2**, on the egress side of a gate **2a**, through which the resin **21** in a molten state is to be injected, a space (resin reservoir **2b**) is formed. The resin **21** injected into the molding device **2** moves via the resin reservoir **2b** in the axial direction of the mandrel **1**. The resin **21** as described above interpenetrates into the first carbon fiber layer **11**, the second carbon fiber layer **12**, and the third carbon fiber layer **13**. In a state where the resin **21** has interpenetrated into the carbon fiber layers **11** to **13**, heat is applied to the molding device **2** and pressure is applied inside the molding device **2**, thereby to form the tube body **20**. In the molding step, the first fixing member **13b** may melt due to the heat(s) of the molding device (mold) **2** and/or of the resin **21** and may mix into the resin **21**.

[0041] Subsequently, the molded tube body **20A** and the mandrel **1** are taken out from the molding device **2** and then the mandrel **1** is pulled out (step S7: Core removal step) from the tube body **20A**. Subsequently, a joint (the other stub yoke or stub shaft) **23** is attached (step S8: Joint attachment step) to the other axial direction end portion of the mandrel **1**.

[0042] The tube body intermediate **10A** according to the first embodiment of the present invention includes: a fiber body (carbon fibers **13a**) disposed with respect to the outer circumferential surface of the mandrel **1** so as to extend in the axial direction of the mandrel **1**; and the first fixing member **13b** wound with respect to outer circumferential surface of the mandrel **1** such that the first fixing member **13b** is wound over the fiber body by one or more turns in the circumferential direction, along the axial direction of the mandrel **1**.

[0043] The tube body production method according to the first embodiment of the present invention includes: a disposing step of disposing the fiber body (carbon fibers **13a**) with respect to the outer circumferential surface of the mandrel **1** so that the fiber body extends in the axial direction of the mandrel **1**; a fixation step of winding the first fixing member **13b**, which is configured to fix the fiber body with respect to the outer circumferential surface of the mandrel **1**, with respect to the outer circumferential surface of the mandrel **1** such that the first fixing member **13b** is wound over the fiber body by one or more turns in the circumferential direction, along the axial direction of the mandrel **1**; and a molding step of impregnating the fiber body with a resin **21** on the outer circumferential surface of the mandrel **1** and then heating the resin **21** to mold the resin **21**.

[0044] These configurations make it possible to, while reducing the production cost, reduce displacement of the fibers even when the orientation angle θ of the fiber body is small.

Second Embodiment

[0045] Subsequently, a description will be given of a tube body intermediate and a tube body production method according to a second embodiment of the present invention, while focusing on differences from the first embodiment.

[0046] As illustrated in FIG. 9, a tube body intermediate **10B** according to the second embodiment further includes a fourth carbon fiber layer **14** disposed on the radially outer side of the third carbon fiber layer **13**.

<<Fourth Carbon Fiber Layer>>

[0047] The fourth carbon fiber layer **14** is disposed on the radially outer side of the third carbon fiber layer **13** and is constituted by: a plurality of carbon fibers **14a** disposed so as to cover the third carbon fiber layer **13**; and a first fixing member **14b**. The carbon fibers **14a** of the fourth carbon fiber layer **14** are disposed to extend in parallel with the axial direction of the mandrel **1**. In other words, the orientation angle θ of the carbon fibers **14a** with respect to the axis X of the mandrel **1** is 0 degrees. The carbon fibers **14a** has a length equal to the axial direction length L of a portion of the mandrel **1** on which portion the carbon fiber layers **11** to **14** are disposed.

<<First Fixing Member>>

[0048] The first fixing member **14b** is a member configured to fix the carbon fibers **14a** on the

outer circumferential surface of the mandrel **1**. The first fixing member **14b** is a flexible resin member having, for example, a thread-like or belt-like shape. The first fixing member **14b** may be formed of the same material as the later-described resin **21**. The first fixing member **14b** may be formed of a material that melts due to the heat(s) of a molding device (mold) **2** and/or of the resin **21** to mix into the resin **21**. The first fixing member **14b** is disposed such that the first fixing member **14b** is wound by one or more turns so as to be inclined with respect to the axial direction of the mandrel **1** and that the first fixing member **14b** extends helically with respect to the axial direction of the mandrel **1**. The first fixing member **14b** is disposed with respect to the first fixing member **13b** at a different phase in the axial direction of the mandrel **1** and in parallel with each other. In other words, the first fixing member **14b** and the first fixing member **13b** are disposed so as not to overlap with each other in the radial direction of the mandrel **1**.

[0049] The first fixing member **14b** as described above prevents the carbon fibers **14a**, which are disposed on the outer circumferential surface of the mandrel **1** placed such that the axial direction thereof extends in the horizontal direction (i.e., which are disposed on the outer circumferential surface of the horizontally placed mandrel **1**), from hanging down due to gravity. Specifically, the first fixing member **14b** suitably prevents, of a portion of carbon fibers **14a** located in a lower portion of the outer circumferential surface of the mandrel **1**, an axial direction intermediate portion from hanging down due to gravity.

Tube Body Production Method

[0050] Subsequently, a description will be given of a method of producing a tube body **20B** (see FIG. **12**) via the tube body intermediate **10B** according to the second embodiment, using the flowchart illustrated in FIG. **10**.

[0051] In this production method, as illustrated in FIG. **11A**, the carbon fibers **13a** of the third carbon fiber layer **13** is formed (step S4: Third carbon fiber layer forming step/Disposing step) on the outer circumferential surface of the second carbon fiber layer **12** (see FIG. **7**) by a not-illustrated device.

[0052] Subsequently, as illustrated in FIG. **11B**, by a not-illustrated device, the first fixing member **13b** is disposed on the outer circumferential surface of the carbon fibers **13a** of the third carbon fiber layer **13**, thereby the carbon fibers **13a** are fixed (step S5: Third carbon fiber layer forming step; Fixation step) with respect to the mandrel **1**.

[0053] Subsequently, as illustrated in FIG. **11C**, carbon fibers **14a** of the fourth carbon fiber layer **14** are formed (step S4B: Fourth carbon fiber layer forming step/Disposing step) on the outer circumferential surface of the third carbon fiber layer **13** by a not-illustrated device.

[0054] Subsequently, as illustrated in FIG. **11D**, by a not-illustrated device, the first fixing member **14b** is disposed on the outer circumferential surface of the carbon fibers **14a** of the fourth carbon fiber layer **14**, thereby the carbon fibers **14a** are fixed (step S5B: Fourth carbon fiber layer forming step, Fixation step) with respect to the mandrel **1**.

[0055] Note that in the example illustrated in FIG. **9**, the carbon fibers **13a** of the carbon fiber layer **13** and the carbon fibers **14a** of the carbon fiber layer **14** are each densely arranged in the circumferential direction and the carbon fibers **14a**, located on the outer side, are disposed in an overlapping manner on the radially outer side of the carbon fiber layer **13a** and the first fixing member **13b**, located on the inner side. In contrast, in the example illustrated in FIGS. **11A** to **11D**, the carbon fibers **13a** of the third carbon fiber layer **13** are disposed at intervals in the circumferential direction and the carbon fibers **14a** of the fourth carbon fiber layer **14** are respectively disposed in the intervals of the carbon fibers **13**. Even in this structure, the carbon fibers **14a** of the fourth carbon fiber layer **14** are disposed on the radially outer side of the first fixing member **13b** of the third carbon fiber layer **13** in an overlapping manner. In this case, in comparison to a case where the carbon fibers **13a** and carbon fibers **14a** are fixed only with the first fixing member **14b** (a case where the first fixing member **13b** is omitted), the number of the carbon fibers **13a** to be fixed by the first fixing member **13b** and the number of the carbon fibers **14a** to be

fixed by the first fixing member **14b** can be reduced, leading to improvement in the degree of freedom in the design of the first fixing members **13b**, **14b**.

[0056] The tube body intermediate **10B** according to the second embodiment of the present invention includes a plurality of fiber layers each including the fiber body and the first fixing member. The plurality of fiber layers (carbon fibers **13a** and carbon fibers **14a**) are disposed in a manner of being stacked in the radial direction of the mandrel **1**. The first fixing member **13b** of the fiber layer **13**, arranged as a lower layer, and the fiber body (carbon fibers **14a**) of the fiber layer **14**, arranged as an upper layer located on top of the lower layer, are at least stacked in the radial direction. The first fixing member **13b** and the first fixing member **14b** respectively included in the plurality of fiber layers are disposed at different phases in the axial direction of the mandrel (and in parallel with each other (with the same orientation angle)). In other words, the first fixing member **14b** and the first fixing member **13b** are disposed so as not to overlap with each other in the radial direction of the mandrel **1**.

[0057] The tube body production method according to the second embodiment of the present invention further includes repeating the disposing step and the fixation step for a plurality of times before the molding step, thereby to form a plurality of fiber layers each including the fiber body and the first fixing member in a manner of being stacked in the radial direction of the mandrel **1**. The first fixing member **13b** of a lower one of the plurality of fiber layers and the fiber body (carbon fibers **14a**) of an upper one of the plurality of fiber layers, located on top of the lower one, are at least stacked in the radial direction. In the repeating the fixation step for the plurality of times, the first fixing members **13b**, **14b** respectively included in the plurality of fiber layers are disposed at different phases in the axial direction of the mandrel **1**.

[0058] These configurations make it possible to reduce displacement of the fiber body even when the orientation angle θ of the fiber body is small. Moreover, in comparison to a case where the first fixing members **13b** and **14b** are disposed at the same phase such that the first fixing member **13b** and **14b** overlap each other in the radial direction, these configurations make it possible to reduce the thickness of the tube body intermediate **10B** and prevent a stress from being concentrated locally.

Third Embodiment

[0059] Subsequently, a description will be given of a tube body intermediate and a tube body production method according to a third embodiment of the present invention, while focusing on differences from the second embodiment.

[0060] As illustrated in FIG. **13**, a tube body intermediate **10C** according to the third embodiment of the present invention includes a pair of second fixing members **15**, **15** as members for fixing the third carbon fiber layer **13**.

<<Second Fixing Member>>

[0061] The second fixing members **15** are members for fixing end portions of the third carbon fiber layer **13** to the outer circumferential surfaces of the axial direction end portions of the mandrel **1**. The second fixing members **15** are each a belt-like flexible resin member having an adhesive part on one surface side (on the radially inner surface side). The second fixing members **15** are each wound in an annular shape. The second fixing members **15** may each be formed of the same material as the resin **21** or may each be formed of a material that melts due to the heat(s) of a molding device (mold) **2** and/or of the resin **21** to mix into the resin **21**, like the first fixing member **14b**.

Tube Body Production Method

[0062] Subsequently, a description will be given of a method of producing a tube body via the tube body intermediate **10C** according to the third embodiment, using the flowchart illustrated in FIG. **14**.

[0063] Between the fourth carbon fiber layer forming step (step **S4B**) and the molding step (step **S6**), the second fixing members **15**, **15** are disposed on the outer circumferential surface of the

opposite end portions of the fourth carbon fiber layer **14** by a not-illustrated device, so that the opposite end portions of the third carbon fiber layer **13** and the opposite end portions of the fourth carbon fiber layer **14** are fixed (step S5C, opposite end portions fixing step) with respect to the mandrel **1**. The fixation by the second fixing members **15**, **15** is preferably performed before the fixation (step S5B) by the first fixing members **14b**, **14b**. Moreover, the fixation by the second fixing members **15**, **15** may be performed between the third carbon fiber layer forming step (step S4) and the fourth carbon fiber layer forming step (step S4B).

[0064] The tube body intermediate **10C** according to the third embodiment of the present invention includes the second fixing members **15** that fix the opposite end portions of the fiber body with respect to the outer circumferential surface of the mandrel **1**.

[0065] According to the tube body production method according to the third embodiment of the present invention, in the fixation step, the opposite end portions of the fiber body is fixed with respect to the outer circumferential surface of the mandrel **1** by the second fixing members **15**.

[0066] These configurations make it possible to, while reducing the production cost, suitably reduce displacement of the fiber body even when the orientation angle θ of the fiber body is small.

[0067] Although certain embodiments of the present invention have been described above, it is to be understood that the present invention is not limited only to the above-described embodiments and the embodiments of the present invention can be modified as appropriate within the range not departing from the gist of the invention. For example, the first carbon fiber layer **11** and the second carbon fiber layer **12** may be omitted and the third carbon fiber layer **13** may be disposed directly on the outer surface of the mandrel **1**. Moreover, the orientation angle of the carbon fibers **13a** of the third carbon fiber layer **13** and the orientation angle of the carbon fibers **14a** of the fourth carbon fiber layer **14** are not limited to 0 degrees (not limited to being in parallel to the axial direction of the mandrel **1**), but may be an angle such that the carbon fibers **13a** and carbon fibers **14a** are wound by less than one turn with respect to the mandrel **1**. In other words, the present invention is preferably applied when the orientation angle θ of the carbon fibers **13a** satisfies $\tan \theta < |2\pi r/L|$. Moreover, the fiber bodies used in the tube body intermediates **10A** to **10C** are not limited to the carbon fibers **11a** to **14a** and may be other materials capable of reinforcing the tube body **20**. Moreover, in the first embodiment, a plurality of the first fixing members **13b** may be disposed at different phases in the axial direction of the mandrel **1** and in parallel with one another. Moreover, in the second embodiment, a plurality of the first fixing members **13b** and a plurality of the first fixing members **14b** may be disposed at different phases in the axial direction of the mandrel **1** and in parallel with one another.

Claims

1. A tube body production method comprising: a step of repeating a fiber layer forming step for a plurality of times to form a plurality of fiber layers so that the plurality of fiber layers are disposed in a manner of being stacked in a radial direction of a circular cylindrical tube member, each fiber layer forming step comprising, to form a respective fiber layer comprised of a respective fiber body and a respective first fixing member, a disposing step of disposing the respective fiber body with respect to an outer circumferential surface of the circular cylindrical tube member so that the respective fiber body extends in an axial direction of the circular cylindrical tube member, and a fixation step of winding the respective first fixing member, which is configured to fix the respective fiber body with respect to the outer circumferential surface of the circular cylindrical tube member, with respect to the outer circumferential surface of the circular cylindrical tube member such that the respective first fixing member is wound over the respective fiber body by one or more turns in a circumferential direction, along the axial direction of the circular cylindrical tube member; and a molding step of, after the step of repeating the fiber layer forming step, impregnating the fiber bodies of the plurality of fiber layers with a resin on the outer

circumferential surface of the circular cylindrical tube member and then heating the resin to mold the resin, wherein the first fixing member of a lower one of the plurality of fiber layers and the fiber body of an upper one of the plurality of fiber layers, located on top of the lower one, are at least stacked in the radial direction, and wherein each of the repeated fixation steps comprises disposing the first fixing member of the corresponding fiber layer at a different phase with respect to the first fixing members disposed in the other fixation steps in the axial direction of the circular cylindrical tube member.

2. The tube body production method according to claim 1, wherein the fixation step of one of the repeated fiber layer forming steps further comprises fixing, by a second fixing member, opposite end portions of the fiber body disposed by the disposing step of the one of the repeated fiber layer forming steps, with respect to the outer circumferential surface of the circular cylindrical tube member.

3. The tube body production method according to claim 1, wherein the circular cylindrical tube member is a mandrel.

4. The tube body production method according to claim 2, wherein the circular cylindrical tube member is a mandrel.
